

Chapter 1

The properties of gases

Summary

- The perfect gas law

$$pV = nRT$$

R : gas constant

$$pV_m = RT$$

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$$

- Dalton's law

The partial pressure of a perfect gas is the pressure that it would exert if it occupied the container alone.

$$p_i = n_i \frac{RT}{V}$$

- The compression factor: Z

$$Z = \frac{V_m}{V_m^\circ}$$

$$pV_m = ZRT$$

Z : compression factor

V_m : the molar volume of a real gas

V_m° : the molar volume of a perfect gas at the same pressure and temperature.

The perfect gases $Z = \frac{pV_m}{RT} = 1$

Real gases

$Z > 1$ $V_m > V_m^\circ$ expansion

$Z < 1$ $V_m < V_m^\circ$ compression

At very low pressures: $Z \approx 1$.

At high pressures: $Z > 1$.

Repulsive forces are now dominant.

At intermediate pressures: $Z < 1$

For most gases, the attractive forces are dominant.

A perfect gas has $Z = 1$ at all pressures.

- The van der Waals gas--- real gas law vs perfect gas law

$$\left(p + \frac{n^2 a}{V^2}\right) (V - nb) = nRT \quad \Rightarrow \quad p = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2}$$

$$\left(p + \frac{a}{V_m^2}\right) (V_m - b) = RT \quad \Rightarrow \quad p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

The constants a and b are the van der Waals coefficients. They are characteristic of each gas but independent of the temperature.

- The principle of corresponding states 对比状态定律

Reduced variables (对比状态):

Dividing the actual variable by the corresponding critical constant

$$p_r = \frac{p}{p_c} \quad , \quad V_r = \frac{V_m}{V_{m,c}} \quad , \quad T_r = \frac{T}{T_c}$$

$$Z = \frac{pV_m}{RT} = \frac{p_c V_{m,c}}{RT_c} \cdot \frac{p_r V_r}{T_r} = Z_c \frac{p_r V_r}{T_r} \quad Z_c = \frac{p_c V_{m,c}}{RT_c} = \frac{3}{8}$$

$$Z_c \approx 0.375$$

If p_r , V_r , T_r are the same, Z is the same

$Z = f(T_r, p_r)$ ——— for all real gases

No calculation requirement

1.6(a) A perfect gas undergoes isothermal compression, which reduces its volume by 2.20L. The final pressure and volume of the gas are $3.78 \times 10^3 \text{ Torr}$ and 4.56L, respectively. Calculate the original pressure of the gas in (a) Torr,(b)atm.

	P	V	T
Initial	?	6.76 L	T
Final	$3.78 \times 10^3 \text{ Torr}$	4.56 L	T

$$p_i V_i = p_f V_f$$

$$p_i \times 6.76 \text{ L} = 3.78 \times 10^3 \text{ Torr} \times 4.56 \text{ L}$$

$$p_i = 2.55 \times 10^3 \text{ Torr} = 3.36 \text{ atm}$$

$$1 \text{ Torr} = 1.31579 \times 10^{-3} \text{ atm}$$

1.13(b) A gas mixture consists of 320 mg of methane, 175 mg of argon, and 225 mg of neon. The partial pressure of neon at 300 K is 8.87 kPa. Calculate (a) the volume and (b) the total pressure of the mixture.

$$n_{Ne} = \frac{0.225g}{20.18g\text{mol}^{-1}} = 0.01115 \text{ mol}$$

$$p_{Ne} = \frac{n_{Ne}RT}{V} \quad \Rightarrow \quad V = \frac{n_{Ne}RT}{p_{Ne}} = \frac{0.01115 \times 8.314 \times 300}{8870} = 3.14 \times 10^{-3} \text{ m}^3$$

$$n_{CH_4} = \frac{0.320g}{16.04g\text{mol}^{-1}} = 0.01995 \text{ mol} \quad n_{Ar} = \frac{0.175g}{39.95g\text{mol}^{-1}} = 0.00438 \text{ mol}$$

$$n = n_{CH_4} + n_{Ne} + n_{Ar} = 0.0355 \text{ mol}$$

$$p = \frac{nRT}{V} = \frac{0.0355 \times 8.314 \times 300}{3.14 \times 10^{-3}} = 28.2 \text{ kPa}$$

1.16(a) Calculate the pressure exerted by 1.0 mol H_2S behaving as (a) a perfect gas, (b) a van der Waals gas when it is confined under the following conditions: (i) at 273.15 K in 22.414 L, (ii) at 500 K in 150 cm^3 . Use the data in Table 1.5.

1.16 (a) as a perfect gas

$$P_1 = \frac{nRT_1}{V_1} = \frac{1 \text{ mol} \times 8.314 \text{ J/mol} \cdot \text{K} \times 273.15 \text{ K}}{22.414 \times 10^{-3} \text{ m}^3} = 1.013 \times 10^5 \text{ Pa}$$

$$P_2 = \frac{nRT_2}{V_2} = \frac{1 \text{ mol} \times 8.314 \text{ J/mol} \cdot \text{K} \times 500 \text{ K}}{150 \times 10^{-6} \text{ m}^3} = 2.8 \times 10^7 \text{ Pa}$$

(b) as a van der Waals gas

For H_2S (Table 1.5) $a = 4.484 \text{ atm} \cdot \text{L}^2 \cdot \text{mol}^{-2}$

$b = 4.34 \times 10^{-2} \text{ L} \cdot \text{mol}^{-1}$

$R = 8.20578 \times 10^{-2} \text{ L atm K}^{-1} \text{ mol}^{-1}$

$$P_1 = \frac{nRT}{V-nb} - \frac{n^2a}{V^2} = \frac{1 \text{ mol} \times 8.20578 \times 10^{-2} \text{ L atm K}^{-1} \text{ mol}^{-1} \times 273.15 \text{ K}}{22.414 \text{ L} - 1 \text{ mol} \times 4.34 \times 10^{-2} \text{ L} \cdot \text{mol}^{-1}} - \frac{(1 \text{ mol})^2 \times 4.484 \text{ atm} \cdot \text{L}^2 \cdot \text{mol}^{-2}}{(22.414 \text{ L})^2}$$

$$= 1 \text{ atm}$$

$$P_2 = \frac{1 \text{ mol} \times 8.20578 \times 10^{-2} \text{ L atm K}^{-1} \text{ mol}^{-1} \times 500 \text{ K}}{0.15 \text{ L} - 1 \text{ mol} \times 4.34 \times 10^{-2} \text{ L} \cdot \text{mol}^{-1}} - \frac{(1 \text{ mol})^2 \times 4.484 \text{ atm} \cdot \text{L}^2 \cdot \text{mol}^{-2}}{(0.15 \text{ L})^2}$$

$$= 185.6 \text{ atm}$$

1.18(b) A gas at 350 K and 12 atm has a molar volume 12 percent larger than that calculated from the perfect gas law. Calculate (a) the compression factor under these conditions and (b) the molar volume of the gas. Which are dominating in the sample, the attractive or the repulsive forces?

The compression factor is

$$Z = \frac{p V_m}{RT} = \frac{V_m}{V_{m,\text{perfect}}}$$

(a) Because $V_m = V_{m,\text{perfect}} + 0.12 V_{m,\text{perfect}} = (1.12) V_{m,\text{perfect}}$, we have $Z = \boxed{1.12}$
Repulsive forces dominate.

(b) The molar volume is

$$V = (1.12) V_{m,\text{perfect}} = (1.12) \times \left(\frac{RT}{p} \right)$$

$$V = (1.12) \times \left(\frac{(0.08206 \text{ L atm K}^{-1} \text{ mol}^{-1}) \times (350 \text{ K})}{12 \text{ atm}} \right) = \boxed{2.7 \text{ L mol}^{-1}}$$

1.19(a) In an industrial process, nitrogen is heated to 500 K at a constant volume of 1.000 m³. The gas enters the container at 300 K and 100 atm. The mass of the gas is 92.4 kg. Use the van der Waals equation to determine the approximate pressure of the gas at its working temperature of 500 K. For nitrogen, $a = 1.352 \text{ L}^2 \text{ atm mol}^{-2}$, $b = 0.0387 \text{ L mol}^{-1}$.

$$1.19 a. \quad R = 8.20578 \times 10^{-2} \text{ L} \cdot \text{atm} \cdot \text{K}^{-1} \text{ mol}^{-1}$$

$$n = \frac{92.4 \times 10^3 \text{ g}}{28 \text{ g/mol}} \approx 3300 \text{ mol}$$

$$a = 1.352 \text{ L}^2 \text{ atm} \cdot \text{mol}^{-2}$$

$$b = 0.0387 \text{ L} \cdot \text{mol}^{-1}$$

$$V = 1 \text{ m}^3 = 10^3 \text{ L}$$

$$T = 500 \text{ K}$$

$$P = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2}$$

$$= \frac{3300 \text{ mol} \times 8.20578 \times 10^{-2} \text{ L} \cdot \text{atm} \cdot \text{K}^{-1} \text{ mol}^{-1} \times 500 \text{ K}}{10^3 \text{ L} - 3300 \text{ mol} \times 0.0387 \text{ L} \cdot \text{mol}^{-1}} - \frac{(3300 \text{ mol})^2 \times 1.352 \text{ L}^2 \cdot \text{atm} \cdot \text{mol}^{-2}}{(10^3 \text{ L})^2}$$

$$= 140.5 \text{ atm}$$

1.23(b) The critical constants of ethane are $p_c = 48.20 \text{ atm}$, $V_c = 148 \text{ cm}^3 \text{ mol}^{-1}$, and $T_c = 305.4 \text{ K}$. Calculate the van der Waals parameters of the gas and estimate the radius of the molecules.

The critical volume of a van der Waals gas is

$$V_c = 3b$$

$$\text{so } b = \frac{1}{3} V_c = \frac{1}{3} (148 \text{ cm}^3 \text{ mol}^{-1}) = 49.3 \text{ cm}^3 \text{ mol}^{-1} = \boxed{0.0493 \text{ L mol}^{-1}}$$

By interpreting b as the excluded volume of a mole of spherical molecules, we can obtain an estimate of molecular size. The centres of spherical particles are excluded from a sphere whose radius is the diameter of those spherical particles (i.e., twice their radius); that volume times the Avogadro constant is the molar excluded volume b

$$b = N_A \left(\frac{4\pi (2r)^3}{3} \right) \quad \text{so} \quad r = \frac{1}{2} \left(\frac{3b}{4\pi N_A} \right)^{1/3}$$

$$r = \frac{1}{2} \left(\frac{3(49.3 \text{ cm}^3 \text{ mol}^{-1})}{4\pi (6.022 \times 10^{23} \text{ mol}^{-1})} \right)^{1/3} = 1.94 \times 10^{-8} \text{ cm} = \boxed{1.94 \times 10^{-10} \text{ m}}$$

The critical pressure is

$$p_c = \frac{a}{27b^2}$$

$$\text{so } a = 27 p_c b^2 = 27 (48.20 \text{ atm}) \times (0.0493 \text{ L mol}^{-1})^2 = \boxed{3.16 \text{ L}^2 \text{ atm mol}^{-2}}$$

But this problem is overdetermined. We have another piece of information

$$T_c = \frac{8a}{27Rb}$$

According to the constants we have already determined, T_c should be

$$T_c = \frac{8(3.16 \text{ L}^2 \text{ atm mol}^{-2})}{27(0.08206 \text{ L atm K}^{-1} \text{ mol}^{-1}) \times (0.0493 \text{ L mol}^{-1})} = 231 \text{ K}$$

However, the reported T_c is 305.4 K, suggesting our computed a/b is about 25 per cent lower than it should be.

Chapter 2

The First Law



Summary

The energies

- The internal energy, U

The first law of thermodynamics

$$\Delta U = q + w$$



- Work (w)

Free expansion

$$w = 0$$

At a constant pressure

$$w = -p_{ex}\Delta V$$

Reversible

$$w = -\int_{V_i}^{V_f} p dV$$

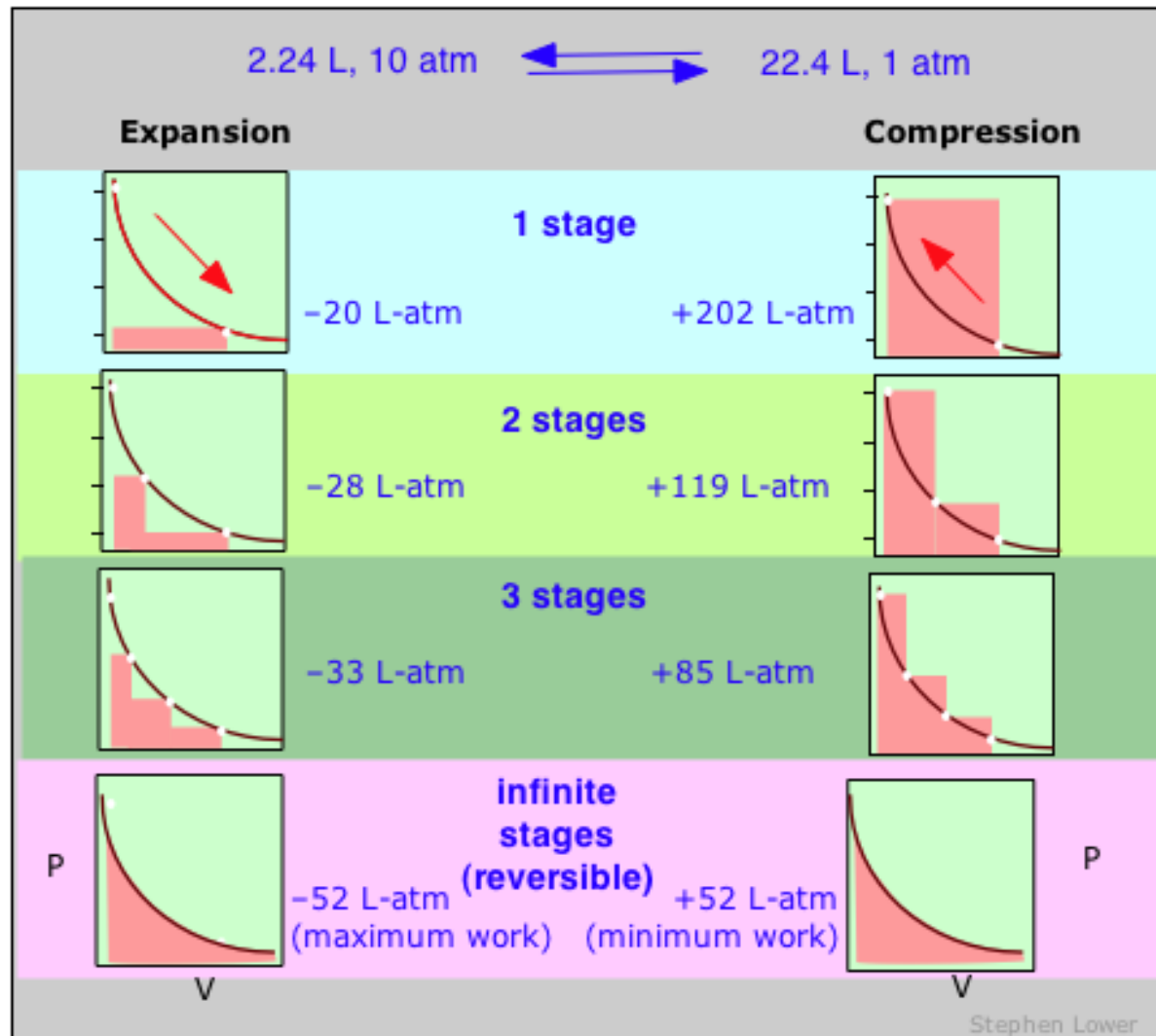
Isothermal reversible

$$w = -nRT \ln \frac{V_f}{V_i}$$

- Heat (q)

$$q_V = \Delta U \text{ (等容)}$$

$$q_p = \Delta H \text{ (等压)}$$





State functions:

Internal energy (U)

$$\Delta U = w + q \quad (1)$$

Enthalpy (H)

$$H = U + PV \quad (2)$$

The equation (2) is always established, no matter whether the pressure is constant or not. But only if the pressure is constant, the following equation (3) can be used.

$$\Delta H = q_p \quad (3)$$

Heat capacity (C)

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V$$

$$C_p = \left(\frac{\partial H}{\partial T} \right)_p$$



Heat capacity of the perfect gases

$$U = f(T); \quad H = f(T)$$

$$C_V = f(T); \quad C_p = f(T)$$

$$C_{p,m} - C_{V,m} = R$$

$$\Delta U = \int C_v dT$$

$$\Delta H = \int C_p dT$$

这两个公式不受限是否等容、等压



Adiabatic reversible process

$q=0$, $\Delta U=w$, p , V , T changing

The change in pressure resulting from an adiabatic, reversible expansion of a perfect gas:

$$V_f T_f^c = V_i T_i^c \quad c = C_{V,m}/R$$

$$p_i V_i^\gamma = p_f V_f^\gamma \quad \gamma = \frac{C_{p,m}}{C_{V,m}}$$

*Real gases: Joule-Tomson effect (no calculation here)

Joule-Tomson coefficient $\mu_{J-T} = (\partial T / \partial p)_H$

- State function
- 等焓过程，压力-温度的变化情况
- $\mu_{J-T} > 0, p \downarrow T \downarrow$ *positive effect*
- $\mu_{J-T} < 0, p \downarrow T \uparrow$ *negative effect*
- For perfect gas, no effect $\mu = 0$

Isenthalps & inversion curve



Thermochemistry

The standard enthalpy change ($\Delta H^\ominus$), the change in enthalpy for a process in which the initial and final substances are in their standard states.

The standard state: of the pure form a substance at 1 bar and at a specified temperature. Conventionally, $T=298.15\text{ K}$ ($25\text{ }^\circ\text{C}$).



$$\Delta_{trs}H^{\ominus}$$

Physical change

$$\Delta_rH^{\ominus}$$

Chemical change

$$\Delta_fH^{\ominus}$$

The standard enthalpy of formation ($\Delta_fH^{\ominus}$)

The standard reaction enthalpy for the formation of the compound from its elements in their reference states.

The reference state of an element is its most stable state at the specified temperature and 1 bar.

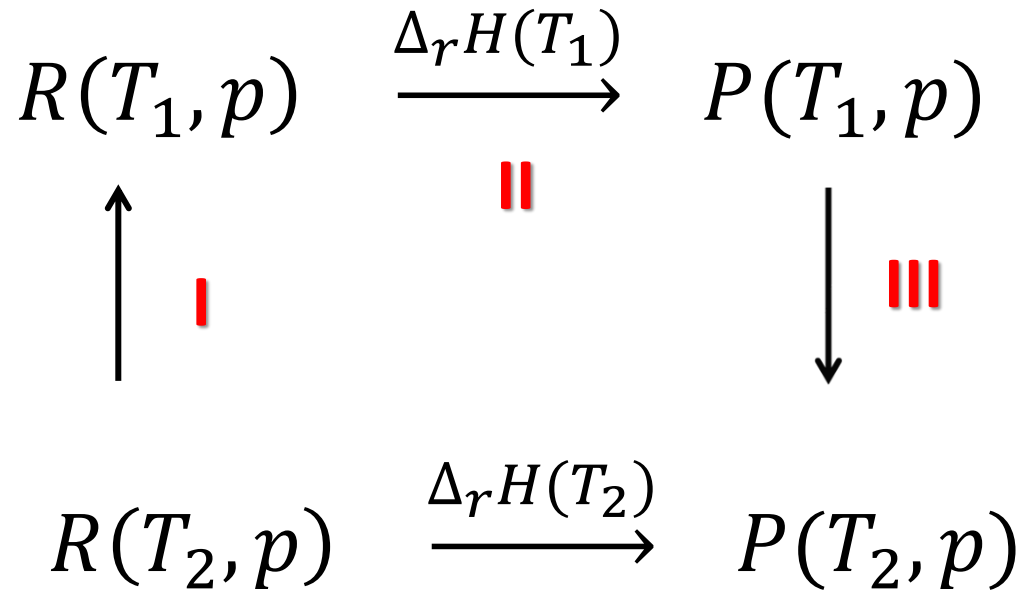


$$\Delta_r H^\ominus = \sum_{\text{Products}} v \Delta H_f^\ominus - \sum_{\text{Reactants}} v \Delta H_f^\ominus$$

$$\Delta_r H^\ominus(T_2) = \Delta_r H^\ominus(T_1) + \int_{T_1}^{T_2} \Delta_r C_p^\ominus dT$$

It is normally a good approximation to assume that $\Delta_r C_p$ is independent of the temperature over reasonably limited ranges.

$$\Delta_r H^\ominus(T_2) = \Delta_r H^\ominus(T_1) + (T_2 - T_1) \Delta_r C_p^\ominus$$



$$\begin{aligned} \Delta_r H(T_2) &= \Delta H_I + \Delta_r H(T_1) + \Delta H_{III} \\ &= \int_{T_2}^{T_1} C_{p,m}(R) dT + \Delta_r H(T_1) + \int_{T_1}^{T_2} C_{p,m}(P) dT \end{aligned}$$

$$\Delta_r H(T_2) = \Delta_r H(T_1) + \int_{T_1}^{T_2} \Delta_r C_{p,m} dT$$

2.6 (a) A chemical reaction takes place in a container of cross-sectional area 100 cm^2 . As a result of the reaction, a piston is pushed out through 10 cm against an external pressure of 1.0 atm . Calculate the work done by the system.

Tell: expansion work under constant p_{ex}

Ask for w

$$w = -p_{ex}\Delta V$$

2.7 (b) A sample consisting of 2.00 mol He is expanded isothermally at 22°C from 22.8 L to 31.7 L (a) reversibly, (b) against a constant external pressure equal to the final pressure of the gas, and (c) freely (against zero external pressure). For the three processes calculate q , w , ΔU , and ΔH .

Isothermal reversible

$$\Delta U = \Delta H = 0$$

$$w = -nRT \ln \frac{V_f}{V_i}$$

$$q = -w$$

Isothermal p_{ex}

$$\Delta U = \Delta H = 0$$

$$p_{ex} = p_f = \frac{nRT}{V_f}$$

$$w = -p_{ex}\Delta V$$

$$q = -w$$

Isothermal *free*

$$\Delta U = \Delta H = 0$$

$$w = 0$$

$$q = 0$$

2.8 (a) A sample consisting of 1.00 mol of monatomic perfect gas, for which $C_{V,m} = \frac{3}{2}R$, initially at $p_1 = 1.00$ atm and $T_1 = 300$ K, is heated reversibly to 400 K at constant volume. Calculate the final pressure, ΔU , q , and w .

At constant volume

$$p_2 = \frac{p_1 T_2}{T_1}$$

$$w = 0$$

$$\Delta U = q_V = nC_{V,m}\Delta T$$

2.9 (a) A sample of 4.50 g of methane occupies 12.7 L at 310 K.
(a) Calculate the work done when the gas expands isothermally against a constant external pressure of 200 Torr until its volume has increased by 3.3 L. (b) Calculate the work that would be done if the same expansion occurred reversibly.

(a)

$$w = -p_{ex}\Delta V$$

(b) Isothermal reversible

$$w = -nRT \ln \frac{V_f}{V_i}$$

2.10 (a) In the isothermal reversible compression of 52.0 mmol of a perfect gas at 260 K, the volume of the gas is reduced to one-third its initial value. Calculate w for this process.

Isothermal reversible

$$w = -nRT \ln \frac{V_f}{V_i}$$

2.14(b) The amount of the perfect gas is 1 mol.

2.14 (b) The constant-pressure heat capacity of a sample of a perfect gas was found to vary with temperature according to the expression $C_p/(\text{J K}^{-1}) = 20.17 + 0.4001(T/\text{K})$. Calculate q , w , ΔU , and ΔH when the temperature is raised from 0°C to 100°C (a) at constant pressure, (b) at constant volume.

(a) Constant pressure

$$\Delta H = \int C_p dT$$
$$\Delta U = q + w$$

$$q = \Delta H \quad w = -p\Delta V = -nR\Delta T$$

(b) Constant volume

因为只是T的函数，这两个值实际没变化

$$\Delta H = \int C_p dT$$
$$\Delta U = \Delta H - V\Delta p$$

$$w = p\Delta V = 0$$

$$q = \Delta U - w = \Delta U$$

(a) At constant pressure, $q = \Delta H$.

$$\begin{aligned} q &= \int C_p dT = \int_{0+273\text{ K}}^{100+273\text{ K}} [20.17 + (0.4001)T/\text{K}] dT \text{ J K}^{-1} \\ &= \left[(20.17) T + \frac{1}{2} (0.4001) \times \left(\frac{T^2}{\text{K}} \right) \right] \bigg|_{273\text{ K}}^{373\text{ K}} \text{ J K}^{-1} \\ &= \left[(20.17) \times (373 - 273) + \frac{1}{2} (0.4001) \times (373^2 - 273^2) \right] \text{ J} = \boxed{14.9 \times 10^3 \text{ J}} = \Delta H \\ w &= -p\Delta V = -nR\Delta T = -(1.00 \text{ mol}) \times (8.3145 \text{ J K}^{-1} \text{ mol}^{-1}) \times (100 \text{ K}) = \boxed{-831 \text{ J}} \\ \Delta U &= q + w = (14.9 - 0.831) \text{ kJ} = \boxed{14.1 \text{ kJ}} \end{aligned}$$

(b) The energy and enthalpy of a perfect gas depend on temperature alone. Thus, $\Delta H = \boxed{14.9 \text{ kJ}}$ and $\Delta U = \boxed{14.1 \text{ kJ}}$ as above. At constant volume, $w = \boxed{0}$ and $\Delta U = q$, so $q = \boxed{+14.1 \text{ kJ}}$.

2.17 (a) Calculate the final pressure of a sample of carbon dioxide that expands reversibly and adiabatically from 57.4 kPa and 1.0 L to a final volume of 2.0 L. Take $\gamma = 1.4$.

The change in pressure resulting from an adiabatic, reversible expansion of a perfect gas:

$$pV^\gamma = \text{constant}$$

2.21 (b) A sample of a liquid of mass 30.5 g is cooled from 288 K to 275 K at constant pressure by the extraction of 2.3 kJ of energy as heat. Calculate q and ΔH and estimate the heat capacity of the sample.

$$q = -2300 \text{ J}$$

At constant pressure

$$\Delta H = q = -2300 \text{ J}$$

$$C_p = \frac{q}{\Delta T}$$

2.23(b) The molar constant-pressure heat capacity of CO_2 is $37.11 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$

2.23 (b) A sample of 5.0 mol CO_2 is originally confined in 15 L at 280 K and then undergoes adiabatic expansion against a constant pressure of 78.5 kPa until the volume has increased by a factor of 4.0 . Calculate q , w , ΔT , ΔU , and ΔH . (The final pressure of the gas is not necessarily 78.5 kPa .)

$$n = 5 \text{ mol} \quad V_1 = 15 \text{ L} \quad T_1 = 280 \text{ K}$$

$$\text{adiabatic} \rightarrow q = 0$$

expansion at a constant pressure

$$w = -p_{\text{ex}}\Delta V = 78.5 \text{ kPa}\Delta V \quad V_2 = 4V_1 = 60 \text{ L}$$

$$q = 0 \quad w = -p_{\text{ex}}\Delta V \quad \Delta U = q + w$$

$$\Delta U = nC_{V,m}\Delta T = w + q = w \rightarrow \Delta T$$

$$\Delta H = \Delta U + \Delta(pV) = \Delta U + nR\Delta T$$

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$$P_1 = \frac{nRT_1}{V_1} = \frac{5 \text{ mol} \times 8.314 \text{ J/mol} \cdot \text{K} \times 280 \text{ K}}{15 \times 10^{-3} \text{ m}^3} = 0.776 \times 10^6 \text{ Pa}$$

$$= 776 \text{ kPa}$$

against a constant pressure $\Rightarrow$ irreversible.

$$W = -785 \times 10^3 \text{ Pa} \times (60 - 15) \times 10^{-3} \text{ m}^3 = -3.53 \times 10^3 \text{ J}$$

$$q = 0$$

$$\Delta U = w + q = -3.53 \times 10^3 \text{ J}$$

$$\Delta U = n \cdot C_{v,m} \times \Delta T = 5 \text{ mol} \times 37.11 \text{ J/mol} \cdot \text{K} \times \Delta T$$

$$= -3.53 \times 10^3 \text{ J}$$

$$\Rightarrow \Delta T = -19 \text{ K}$$

$$\Rightarrow \Delta H = \Delta U + \Delta(PV) = \Delta U + nR\Delta T = -3.53 \times 10^3 \text{ J} +$$

$$5 \text{ mol} \times 8.314 \text{ J/mol} \cdot \text{K} \times (-19 \text{ K})$$

$$= -4.32 \times 10^3 \text{ J}$$

$$\Rightarrow \Delta H = n(C_{v,m} + R) \times \Delta T$$

$$= 5 \text{ mol} \times (37.11 + 8.314) \text{ J/mol} \cdot \text{K} \times (-19 \text{ K})$$

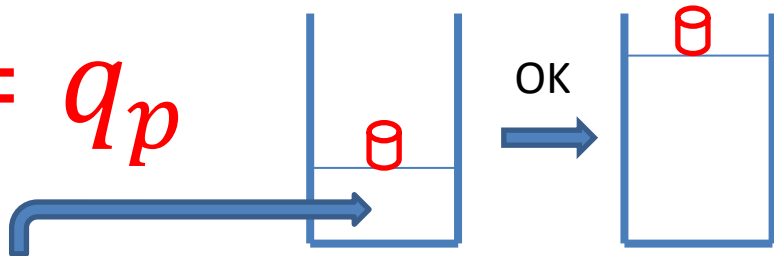
$$= -4.32 \times 10^3 \text{ J}$$

$$H = U + pV$$

Universal, no conditions

$$\Delta H = \Delta U + \Delta(pV) = \Delta U + nR\Delta T$$

$$\Delta H = q_p$$

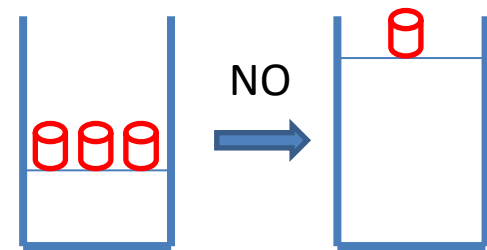


Why the system will expand?
Heating

Initial state
 $p_{ex} = p$

Final state
 $p_{ex} = p$

During the whole process from initial to final state, the internal pressure keeps



Initial state
Final state
 $p_{ex} \neq p$

Although expand against a constant external pressure, The equation is not established.

$$pV^\gamma = \text{constant}$$

Adiabatically reversible expansion

$$dw = -p_{ex}dV$$

External pressure only appear in this equation. In other equations, you see p which will be the pressure of system.

(2.24 (b)) A sample consisting of 2.5 mol of gas molecules at 220 K and 200 kPa is compressed reversibly and adiabatically until the temperature reaches 255 K. Given that its molar constant-volume heat capacity is $27.6 \text{ J K}^{-1} \text{ mol}^{-1}$, calculate q , w , ΔU , ΔH , and the final pressure and volume.

$$n = 2.5 \text{ mol} \quad p_1 = 200 \text{ kPa} \quad T_1 = 220 \text{ K} \quad T_2 = 255 \text{ K}$$

$$\text{adiabatic and reversible} \rightarrow q = 0$$

$$C_{V,m} = 27.6 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$q = 0 \quad \Delta U = nC_{V,m}\Delta T = w + q = w$$

$$\Delta H = \Delta U + \Delta(pV) = \Delta U + nR\Delta T$$

$$V_1 = \frac{nRT_1}{p_1} \quad \Rightarrow \quad V_1$$

$$V_2 T_2^c = V_1 T_1^c \quad c = C_{V,m}/R \quad \Rightarrow \quad V_2$$

The molar constant-pressure heat capacity of O_2 is $29.355 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$

2.27 (b) Consider a system consisting of 3.0 mol O_2 (assumed to be a perfect gas) at 25°C confined to a cylinder of cross-section 22 cm^2 at 820 kPa. The gas is allowed to expand adiabatically and irreversibly against a constant pressure of 110 kPa. Calculate w , q , ΔU , ΔH , and ΔT when the piston has moved 15 cm.

$$n = 3 \text{ mol} \quad p_1 = 820 \text{ kPa} \quad T_1 = 298 \text{ K}$$

adiabatic and irreversible $\rightarrow q = 0$

expansion at a constant pressure

$$w = -p_{\text{ex}}\Delta V = 110 \text{ kPa} \times 22 \text{ cm}^2 \times 15 \text{ cm}$$

$$q = 0 \quad \Delta U = w + q$$

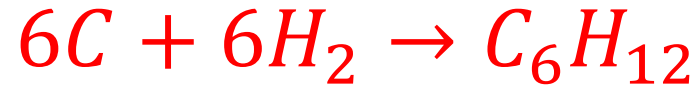
$$\Delta U = nC_{V,m}\Delta T \rightarrow \Delta T$$

$$\Delta H = nC_{p,m}\Delta T \quad \Delta H = \Delta U + \Delta(pV) = \Delta U + nR\Delta T$$



Self-test2.9

Estimate the standard enthalpy of formation of cyclohexane at 400K from the data in Table 2.5 at the end of book.

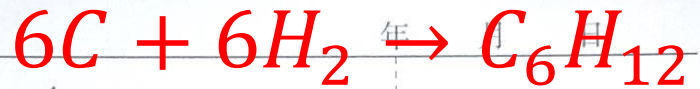


$$\Delta_r H^\ominus(T_2) = \Delta_r H^\ominus(T_1) + \int_{T_1}^{T_2} \Delta_r C_p^\ominus dT$$

$$\Delta_f H^\ominus(T_2) = \Delta_f H^\ominus(298K) + \int_{298}^{T_2} \Delta_r C_p^\ominus dT$$



For C_6H_{12}



$$\Delta_f H^\ominus(298K) = -156 \text{ kJ/mol}$$

$$C_{p,m}^\ominus(C) = 8.527 \text{ J/K}\cdot\text{mol} \quad C_{p,m}^\ominus(H_2) = 28.824 \text{ J/K}\cdot\text{mol}$$

$$C_{p,m}^\ominus(C_6H_{12}) = 156.5 \text{ J/K}\cdot\text{mol}$$

$$i) \Delta_r C_p^\ominus = C_{p,m}^\ominus(C_6H_{12}) - \{6 \times C_{p,m}^\ominus(C) + 6 \times C_{p,m}^\ominus(H_2)\}$$

$$= 156.5 \text{ J/K}\cdot\text{mol} - (6 \times 8.527 + 6 \times 28.824) \text{ J/K}\cdot\text{mol}$$

$$= -67.606 \text{ J/K}\cdot\text{mol}$$

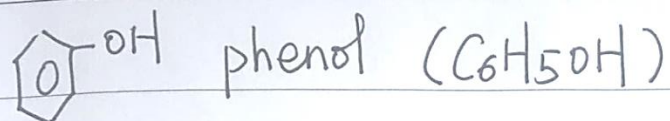
$$\Delta_f H^\ominus(400K) = \Delta_f H^\ominus(298K) + \int_{298}^{400} -67.606 \times 10^{-3} dT$$

$$= -156 \text{ kJ/mol} - (6.76 \times 10^{-2} \times 102) \text{ kJ/mol}$$

$$\Delta_f H^\ominus(400K) = -162.9 \text{ kJ/mol}$$



2.30 (B) The standard enthalpy of formation of phenol is $-165.0 \text{ kJ mol}^{-1}$. Calculate its standard enthalpy of combustion.



Combustion:



$$\Delta_f H^\ominus(\text{C}_6\text{H}_5\text{OH}) = -165.0 \text{ kJ/mol}$$

$$\Delta_f H^\ominus(\text{O}_2, \text{g}) = 0$$

$$\Delta_f H^\ominus(\text{CO}_2, \text{g}) = -393.51 \text{ kJ/mol}$$

$$\Delta_f H^\ominus(\text{H}_2\text{O}, \text{l}) = -285.83 \text{ kJ/mol}$$

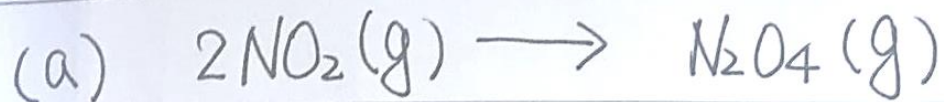
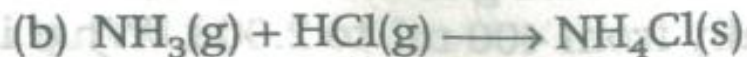
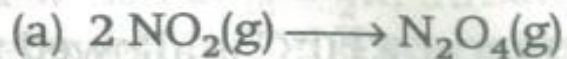
$$\Delta_r H^\ominus = 6\Delta_f H^\ominus(\text{CO}_2) + 3\Delta_f H^\ominus(\text{H}_2\text{O}) - \left\{ \Delta_f H^\ominus(\text{C}_6\text{H}_5\text{OH}) + 7\Delta_f H^\ominus(\text{O}_2) \right\}$$

$$= 6 \times (-393.51) + 3 \times (-285.83) - (-165.0)$$

$$\Delta_r H^\ominus(\text{combustion}) = -3053.55 \text{ kJ/mol}$$



2.39 (a) Use standard enthalpies of formation to calculate the standard enthalpies of the following reactions:



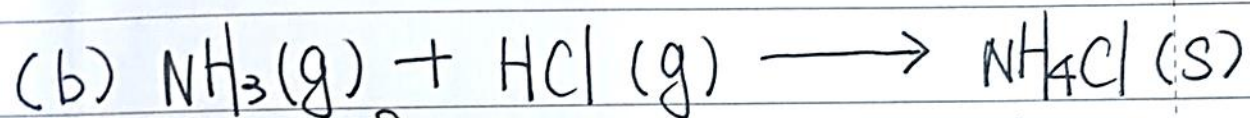
$$\Delta_f H^\ominus(\text{NO}_2, \text{g}) = 33.18 \text{ kJ/mol}$$

$$\Delta_f H^\ominus(\text{N}_2\text{O}_4, \text{g}) = 9.16 \text{ kJ/mol}$$

$$\Delta_r H^\ominus = \Delta_f H^\ominus(\text{N}_2\text{O}_4, \text{g}) - 2 \times \Delta_f H^\ominus(\text{NO}_2, \text{g})$$

$$= 9.16 - 2 \times (33.18)$$

$$= -57.2 \text{ kJ/mol}$$



$$\Delta_f H^\ominus(\text{NH}_3, g) = \cancel{294.1} \text{ kJ/mol} - 46.11$$

$$\Delta_f H^\ominus(\text{HCl}, g) = -92.31 \text{ kJ/mol}$$

$$\Delta_f H^\ominus(\text{NH}_4\text{Cl}, s) = -314.43 \text{ kJ/mol}$$

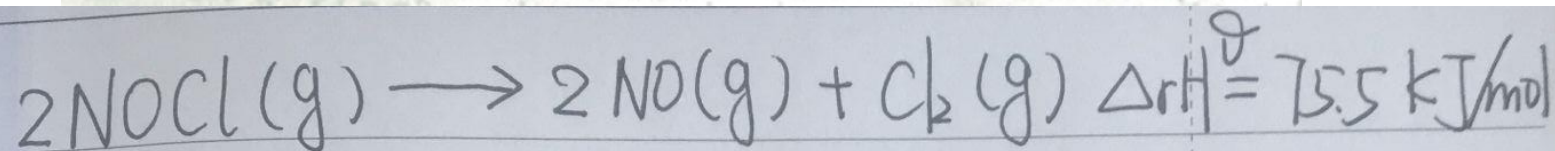
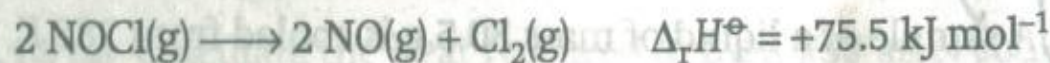
$$\Delta_r H^\ominus = \Delta_f H^\ominus(\text{NH}_4\text{Cl}, s) - \Delta_f H^\ominus(\text{NH}_3, g) - \Delta_f H^\ominus(\text{HCl}, g)$$

$$= -314.43 - (\cancel{294.1} - 46.11) - (-92.31)$$

$$= \cancel{-516.22} \text{ kJ/mol} - 176.01 \text{ kJ/mol}$$



2.42(b) Calculate the standard enthalpy of formation of NOCl(g) from the enthalpy of formation of NO given in Table 2.5, together with the following information:



$$\Delta_f H^\ominus(\text{NO, g}) = 90.25 \text{ kJ/mol}$$

$$\begin{aligned} \Delta_r H^\ominus &= 2 \times \Delta_f H^\ominus(\text{NO, g}) + \Delta_f H^\ominus(\text{Cl}_2, \text{g}) - 2 \times \Delta_f H^\ominus(\text{NOCl, g}) \\ &= 2 \times \Delta_f H^\ominus(\text{NO, g}) - 2 \times \Delta_f H^\ominus(\text{NOCl, g}) \end{aligned}$$

$$75.5 = 2 \times 90.25 - 2 \times \Delta_f H^\ominus(\text{NOCl, g})$$

$$\Delta_f H^\ominus(\text{NOCl, g}) = 52.5 \text{ kJ/mol}$$

1. A sample consisting of 3.7 moles of ideal gas molecules with $C_{p,m} = 17.5 \text{ J K}^{-1} \text{ mol}^{-1}$ is initially at 173 kPa and 323 K. It then undergoes a reversible adiabatic expansion until its temperature is 247 K. (15 points)

1) What is the final volume and pressure?

2) What is heat and work for this process?

$n = 3.7 \text{ mol}$ $C_{p,m} = 17.5 \text{ J/K} \cdot \text{mol}$
 $P_i = 173000 \text{ Pa}$ $T_i = 323 \text{ K}$ $T_f = 247 \text{ K}$ reversible adiabatic expansion.

1. P_f and V_f .
 $V_i = \frac{n \cdot R \cdot T_i}{P_i} = \frac{3.7 \text{ mol} \times 8.314 \text{ J/K} \cdot \text{mol} \times 323 \text{ K}}{173000 \text{ Pa}} = 0.057 \text{ m}^3$
 $V_f T_f^c = V_i T_i^c$ $c = \frac{C_{v,m}}{R} = \frac{C_{p,m} - R}{R} = \frac{17.5 - 8.314}{8.314} = 1.1$
 $V_f \times 247^{1.1} = 0.057 \times (323)^{1.1} = 0.075 \text{ m}^3$
 $V_f = 0.075 \text{ m}^3$
 $P_f = \frac{n R T_f}{V_f} = \frac{3.7 \text{ mol} \times 8.314 \text{ J/mol} \cdot \text{K} \times 247 \text{ K}}{0.075 \text{ m}^3} = 101.3 \text{ kPa}$

2. $\therefore$ adiabatic process. $q = 0$
 $w = \Delta U = n \cdot C_{v,m} \cdot \Delta T = 3.7 \text{ mol} \times (17.5 - 8.314) \text{ J/K} \cdot \text{mol} \times (247 - 323) \text{ K}$
 $w = -2583.1 \text{ J}$

* 气体方程
 * 绝热可逆过程

A sample of 1.0 mol perfect gas is initially at 500 K, 1.0 MPa state. It expands adiabatically against a constant 0.1 MPa external pressure until it reaches balance with the surroundings. Then, it is heated at constant volume until its temperature rises to 500 K. Calculate w , q , ΔU and ΔH for the whole process, given that $C_{V,m} = 20.786 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$. (15 points).

2. $\boxed{n=1 \text{ mol}$
 $T_i=500 \text{ K} \quad P_i=1 \times 10^6 \text{ Pa}$

adiabatically
expansion
 $P_{\text{ex}} = 10^5 \text{ Pa}$

$\boxed{V}$
 $P_f = 10^5 \text{ Pa}$

heated
 $V = \text{constant}$

$\boxed{V}$
 $T = 500 \text{ K}$

$C_{V,m} = 20.786 \text{ J/K} \cdot \text{mol}$

$P_i V_i = n R T_i$

$V_i = \frac{n R T_i}{P_i} = \frac{1 \text{ mol} \times 8.314 \text{ J/K} \cdot \text{mol} \times 500 \text{ K}}{10^6 \text{ Pa}}$
 $= 4.157 \times 10^{-3} \text{ m}^3$

$\Delta U_1 = w_1 + q_1 = w_1 = -P_{\text{ex}} \Delta V_1 = n C_{V,m} \Delta T_1$

$P_2 (V_i + \Delta V_1) = n R (T_i + \Delta T_1)$

$P_2 = P_{\text{ex}} = 10^5 \text{ Pa}$

$\begin{cases} 10^5 \text{ Pa} \times \Delta V_1 = 1 \text{ mol} \times 20.786 \text{ J/K} \cdot \text{mol} \times \Delta T_1 \\ 10^5 \text{ Pa} (4.157 \times 10^{-3} + \Delta V_1) = 1 \text{ mol} \times 8.314 \text{ J/K} \cdot \text{mol} \times (500 + \Delta T_1) \end{cases}$

$\Delta T_1 = -128 \text{ K}$

$w_1 = \Delta U_1 = n C_{V,m} \Delta T_1 = 1 \text{ mol} \times 20.786 \text{ J/K} \cdot \text{mol} \times (-128 \text{ K}) = -2660 \text{ J}$

$w = w_1 + w_2 (=0) = w_1 = -2660 \text{ J}$

$\Delta U = 0 = w$

$q = q_1 + q_2 = q_2$

$\Delta U = \Delta U_1 + \Delta U_2 = w_1 + q_1 + w_2 + q_2 = w_1 + q_2 = 0$

$q_2 = -w_1 = 2660 \text{ J}$

$$\begin{array}{l} n = 1 \text{ mol} \quad V_i \\ T_i = 500 \text{ K} \quad P_i = 1 \times 10^6 \text{ Pa} \end{array}$$

在 $P_{ex} = 1 \times 10^5 \text{ Pa}$

 作用下绝热膨胀

$$\begin{array}{l} n = 1 \text{ mol} \quad V_f \\ T_2 = ? \quad P_2 = 1 \times 10^5 \text{ Pa} \end{array}$$

恒体积加热


$$\begin{array}{l} n = 1 \text{ mol} \quad V_f \\ T_f = 500 \text{ K} \quad P_f = ? \end{array}$$

$$P_i V_i = n R T_i \quad V_i = \frac{n R T_i}{P_i} = \frac{1 \text{ mol} \times 8.314 \text{ J/Kmol} \times 500 \text{ K}}{1 \times 10^6 \text{ Pa}} = 4.157 \times 10^{-3} \text{ m}^3$$

过程a: $\Delta U_1 = w_1 + q_1 = w_1 = -P_{ex} \times \Delta V_a = n \times C_{V,m} \times \Delta T_a$ ①

绝热过程 $q_1=0$

$$P_2 \times (V_i + \Delta V_a) = 1 \text{ mol} \times 8.314 \text{ J/Kmol} \times (500 \text{ K} + \Delta T_a)$$
 ②

过程a结束后建立气体状态方程

方程①&②联立二元二次方程  $\Delta T_a = -128 \text{ K}$

$$\Delta U_1 = w_1 = n \times C_{V,m} \times \Delta T_a = -2660 \text{ J}$$

过程a+b是一个温度不变的总过程，所以对于理想气体内能不变

$$\Delta U = w_1 + q_1 + w_2 + q_2 = w_1 + q_2 \quad q_2 = -w_1 = 2660 \text{ J}$$

$$w = w_1 + w_2 = w_1 = -2660 \text{ J} \quad q = q_1 + q_2 = q_2 = 2660 \text{ J}$$

$$\Delta U = 0; \Delta H = 0$$

1. Carbon dioxide (CO₂) and H₂O can be produced from the reaction between CH₄ and O₂. Answer the following questions using data given below.

1) Calculate the standard enthalpy of the reaction at 298K under 1 bar (6 points).

2) Figure out the molar reaction enthalpy at the temperature of 500 K. (9 points)

$$\Delta_f H^\ominus(\text{CO}, \text{g}) = -110.53 \text{ kJ mol}^{-1}$$

$$\Delta_f H^\ominus(\text{CO}_2, \text{g}) = -393.51 \text{ kJ mol}^{-1}$$

$$\Delta_f H^\ominus(\text{CH}_4, \text{g}) = -74.81 \text{ kJ mol}^{-1}$$

$$\Delta_f H^\ominus(\text{H}_2\text{O}, \text{l}) = -285.83 \text{ kJ mol}^{-1}$$

$$C_{p,m}(\text{CH}_4, \text{g}) = 35.31 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$C_{p,m}(\text{CO}_2, \text{g}) = 37.11 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$C_{p,m}(\text{O}_2, \text{g}) = 36.16 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$C_{p,m}(\text{CO}, \text{g}) = 29.14 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$C_{p,m}(\text{H}_2\text{O}, \text{l}) = 75.291 \text{ J K}^{-1} \text{ mol}^{-1}$$

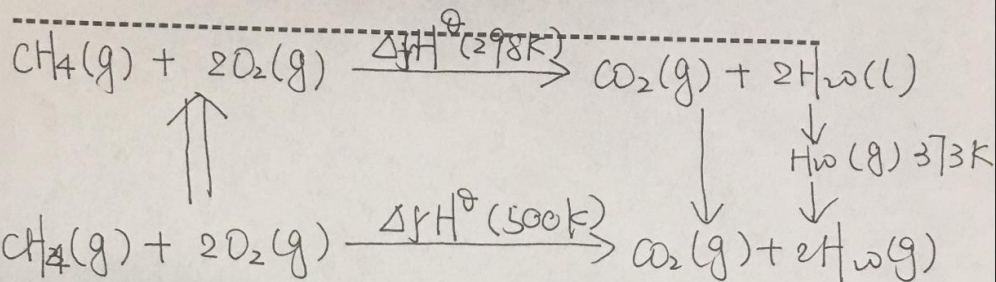
$$C_{p,m}(\text{H}_2\text{O}, \text{g}) = 33.58 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$\Delta_{\text{vap}} H^\ominus(\text{H}_2\text{O}) = 40.656 \text{ kJ mol}^{-1}$$

草

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$$\begin{aligned}
 (1) \Delta_r H^\ominus(298\text{K}) &= \Delta_f H^\ominus(\text{CO}_2, \text{g}) + 2 \times \Delta_f H^\ominus(\text{H}_2\text{O}, \text{l}) - \Delta_f H^\ominus(\text{CH}_4, \text{g}) \\
 &= -393.51 + 2 \times (-285.83) - (-74.81) \\
 &= \underline{-890.36 \text{ kJ/mol}}
 \end{aligned}$$

$$\begin{aligned}
 (2) \Delta_r H^\ominus(500\text{K}) &= C_{p,m}(\text{CH}_4, \text{g})(298-500) + 2 \times C_{p,m}(\text{O}_2, \text{g})(298-500) \\
 &\quad + \Delta_r H^\ominus(298\text{K}) + C_{p,m}(\text{CO}_2, \text{g})(500-298) + 2 \times C_{p,m}(\text{H}_2\text{O}, \text{l})(373-298) \\
 &\quad + 2\Delta_{\text{vap}} H^\ominus(\text{H}_2\text{O}) + 2C_{p,m}(\text{H}_2\text{O}, \text{g})(500-373) \\
 &= 35.31 \times (-202) + 2 \times 36.16 \times (-202) + (-890.36 \times 10^3) + 37.11 \times 202 \\
 &\quad + 2 \times 75.291 \times 75 + 2 \times 40656 + 2 \times 33.58 \times 127 \\
 &= \underline{-803.5 \text{ kJ}}
 \end{aligned}$$

Chapter 3

The Second Law



Summary

Reversible $\longleftrightarrow$

Equilibrium

Irreversible $\longleftrightarrow$

Spontaneous
Non-spontaneous

q

Conditional



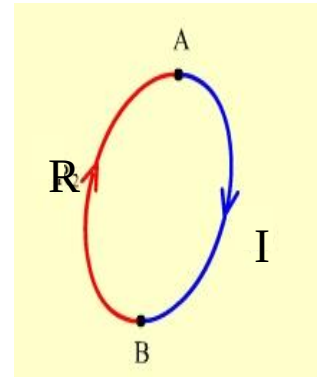
Unconditional

w



S (entropy): State function $J \cdot K^{-1}$

$$dS = \frac{dq_{rev}}{T} \quad \Delta S = \int \frac{dq_{rev}}{T}$$



$$\text{Carnot: } \varepsilon = \frac{w}{q_h} = 1 + \frac{q_c}{q_h}; \varepsilon_{rev} = 1 - \frac{T_c}{T_h}$$

$\varepsilon \leq \varepsilon_{rev}$: = reversible; < spontaneous

$$\text{Clausius inequality: } \Delta S_{A \rightarrow B} \geq \left(\sum_A^B \frac{\delta q}{T} \right)$$

= reversible; > spontaneous

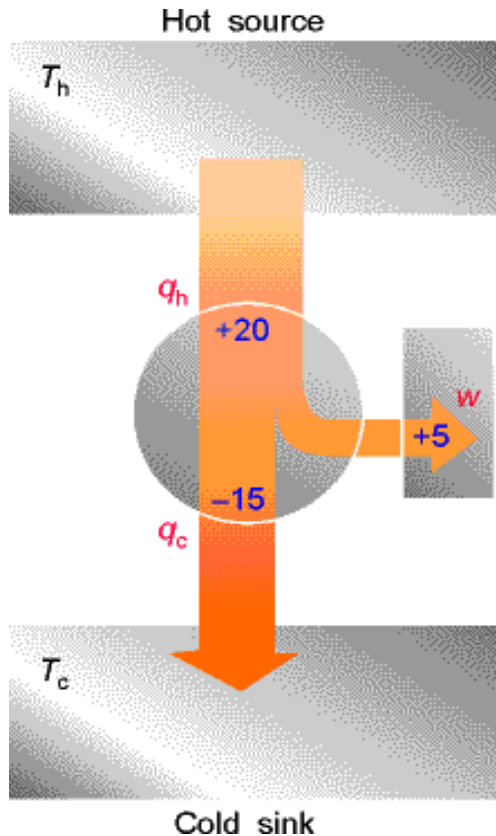
$$\text{Adiabatic system: } q=0 \rightarrow \Delta S_{A \rightarrow B} \geq 0$$

= reversible; > irreversible

$$\text{Isolated system: } \Delta S_{A \rightarrow B} \geq 0$$

= reversible; > spontaneous

- The efficiency of a heat engine (P4.32)



The structure of a heat engine

$$\varepsilon = \frac{\text{work performed}}{\text{heat adsorbed}} = \frac{|w|}{q_h}$$

$$\varepsilon = \frac{q_h + q_c}{q_h} = 1 + \frac{q_c}{q_h} \quad q_c < 0$$

$$w = -(q_h + q_c)$$

The efficiency is the work done divided by the heat supplied from the hot source.



For a Carnot cycle:

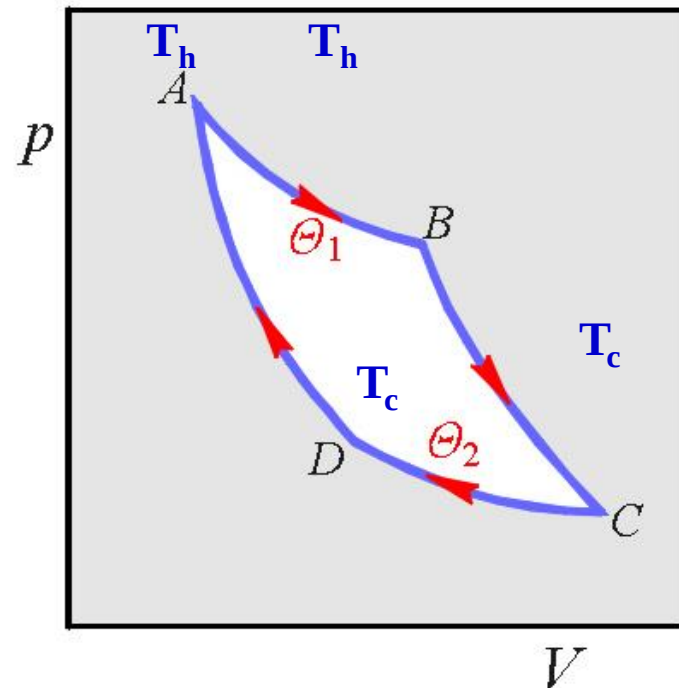
$A \rightarrow B$: isothermal reversible expansion

$B \rightarrow C$: adiabatic reversible expansion

$C \rightarrow D$: isothermal reversible compression

$D \rightarrow A$: adiabatic reversible compression

$$\varepsilon = \frac{q_h + q_c}{q_h} = 1 + \frac{q_c}{q_h} = 1 - \frac{T_c}{T_h}$$





A→B: isothermal reversible expansion

For perfect gas, $U=U(T)$, $q_h+w=0$

$$q_h = -w = nRT_h \ln(V_B/V_A) \quad (1)$$

B→C: adiabatic reversible expansion

$$V_C T_c^c = V_B T_h^c \quad (2)$$

C→D: isothermal reversible compression

For perfect gas, $U=U(T)$, $q_c+w=0$

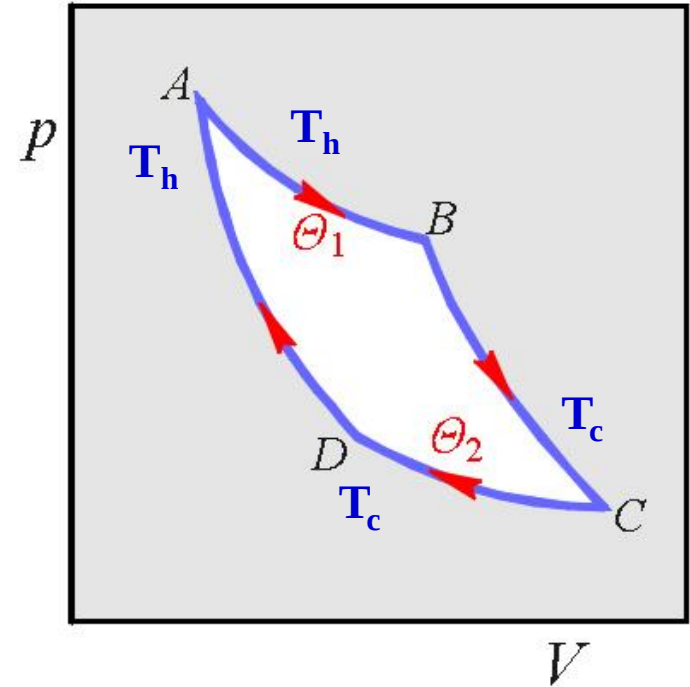
$$q_c = -w = nRT_c \ln(V_D/V_C) \quad (3)$$

D→A: adiabatic reversible compression

$$V_A T_h^c = V_D T_c^c \quad (4)$$

$$(2) \times (4) \longrightarrow V_A V_C T_h^c T_c^c = V_D V_B T_h^c T_c^c$$

$$\longrightarrow V_A V_C = V_D V_B$$



$$\frac{q_h}{q_c} = \frac{nRT_h \ln(V_B/V_A)}{-nRT_c \ln(V_D/V_C)} = -\frac{T_h}{T_c}$$



- Calculate ΔS

相平衡

$$\Delta_{trs}S = \frac{\Delta_{trs}H}{T_{trs}}$$

恒温可逆(or 可应用于不可逆)膨胀

$$\Delta S = nR \ln \frac{V_f}{V_i} = nR \ln \frac{p_i}{p_f}$$

等压变温

$$\Delta S = C_p \ln \frac{T_f}{T_i}$$

等容变温

$$\Delta S = C_V \ln \frac{T_f}{T_i}$$



- Calculate surrounding ΔS_{sur}

The change in entropy of the surroundings

$$dS_{\text{sur}} = \frac{dq_{\text{sur,rev}}}{T_{\text{sur}}}$$

Because the temperature and pressure of the surroundings is commonly constant, whatever the change takes place, reversibly or irreversibly, we have

$$dS_{\text{sur}} = \frac{dq_{\text{sur}}}{T_{\text{sur}}}$$
$$\Delta S_{\text{sur}} = \frac{q_{\text{sur}}}{T_{\text{sur}}} = \frac{-q}{T_{\text{sur}}}$$



- Third law of thermodynamics

$$S(0\text{ K}) = 0$$

The Third-Law entropy, standard entropy $S^\ominus(T)$

The standard reaction entropy $\Delta_r S^\ominus$

$$\Delta_r S^\ominus = \sum_{\text{Products}} \nu S^\ominus - \sum_{\text{Reactants}} \nu S^\ominus$$



- **Helmholtz function**

At constant temperature and volume

$$A = U - TS$$

$$\Delta A \leq w \quad w: \text{non-expansion work}$$

$$w_{max} = \Delta A \quad \text{work function}$$

No non-expansion work

$$dA_{T,V} \leq 0 \quad \left\{ \begin{array}{l} < 0 \text{ spontaneous} \\ = 0 \text{ equilibriums} \end{array} \right.$$

Surrounding entropy change System entropy change

$$dA = \boxed{dU} - \boxed{TdS} \leq 0$$

- **Gibbs Energy**

At constant temperature and pressure

$$G = H - TS$$

$$\Delta G \leq w \quad w: \text{non-expansion work}$$

$$w_{max} = \Delta G \quad \text{work function}$$

No non-expansion work

$$dG_{T,p} \leq 0 \quad \left\{ \begin{array}{l} < 0 \text{ spontaneous} \\ = 0 \text{ equilibriums} \end{array} \right.$$

Surrounding entropy change System entropy change

$$dG = \boxed{dH} - \boxed{TdS} \leq 0$$



The enthalpy of a perfect gas is related to its internal energy by using $pV = nRT$ in the definition of H :

$$H = U + pV = U + nRT$$

This relation implies that the change of enthalpy in reaction that produces or consumes gas is:

$$\Delta_r H = \Delta_r U + \Delta n_g RT$$

(P2.8)



Five state functions in the First and Second Laws

The internal energy: U

The entropy: S

The enthalpy: H

The Helmholtz energy: A

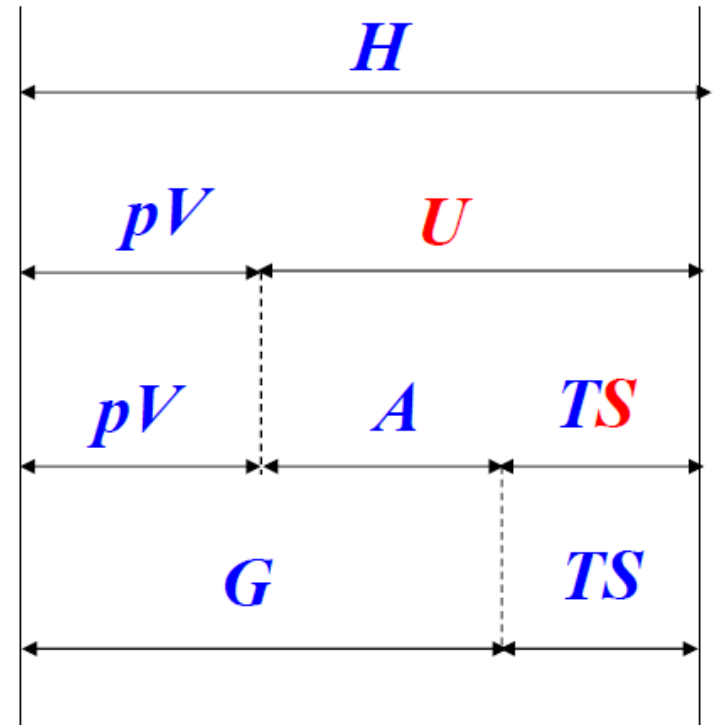
The Gibbs energy: G

$$H = U + PV$$

$$G = A + PV$$

$$A = U - TS$$

$$G = H - TS$$





$$dU = TdS - pdV$$

$$dH = TdS + Vdp$$

$$dG = Vdp - SdT$$

$$dA = -pdV - SdT$$



$$dU = TdS - pdV$$

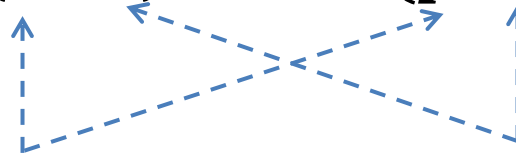


Thermal
Equilibrium

Mechanical
Equilibrium

(T, S)

(p, V)



Intensive properties

Extensive properties



- The Maxwell relations

$$dz = gdx + hdy$$

$$g = (\partial z / \partial x)_y$$

$$h = (\partial z / \partial y)_x$$

$$\left(\frac{\partial g}{\partial y} \right)_x = \left(\frac{\partial h}{\partial x} \right)_y$$



$$dU = TdS - pdV$$

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V$$

$$\left(\frac{\partial U}{\partial S}\right)_V = T, \quad \left(\frac{\partial U}{\partial V}\right)_S = -p$$

$$dH = TdS + Vdp$$

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p$$

$$\left(\frac{\partial H}{\partial S}\right)_p = T, \quad \left(\frac{\partial H}{\partial p}\right)_S = V$$

$$dA = -p dV - SdT$$

$$\left(\frac{\partial p}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T$$

$$\left(\frac{\partial A}{\partial V}\right)_T = -p, \quad \left(\frac{\partial A}{\partial T}\right)_V = -S$$

$$dG = Vdp - SdT$$

$$\left(\frac{\partial V}{\partial T}\right)_p = -\left(\frac{\partial S}{\partial p}\right)_T$$

$$\left(\frac{\partial G}{\partial p}\right)_T = V, \quad \left(\frac{\partial G}{\partial T}\right)_p = -S$$



- Properties of the Gibbs energy

$$G = f(T, p)$$

$$dG = Vdp - SdT$$

$$\left(\frac{\partial G}{\partial p}\right)_T = V \qquad \left(\frac{\partial G}{\partial T}\right)_p = -S$$



- The Gibbs-Helmholtz ($G-H$) equation

$$\left(\frac{\partial}{\partial T} \left(\frac{\Delta G}{T} \right) \right)_p = - \frac{\Delta H}{T^2} \quad \Rightarrow \quad \boxed{\frac{\Delta G_2}{T_2} - \frac{\Delta G_1}{T_1} = - \int_{T_1}^{T_2} \frac{\Delta H}{T^2} dT}$$

$$H = \left(\frac{\partial (G/T)}{\partial (1/T)} \right)_p$$

It shows that if we know the enthalpy of the system, then we know how G/T varies with temperature.



- The variation of the Gibbs energy with pressure

Liquids and solids

$$\Delta G_m = V_m \Delta p$$

Gases

$$\Delta G_m = RT \ln \frac{p_f}{p_i}$$

$$G(p) = G^\ominus + nRT \ln \left(\frac{p}{p^\ominus} \right)$$



- The standard Gibbs energies of reaction**

Standard Gibbs energies of formation $\Delta_f G^\ominus$

The standard Gibbs energy of reaction $\Delta_r G^\ominus$

$$\Delta_r G^\ominus = \sum_{\text{Products}} v \Delta_f G^\ominus - \sum_{\text{Reactants}} v \Delta_f G^\ominus$$



- Real gas and fugacity (no calculation)

$$\varphi = \frac{f}{p}$$

φ : fugacity coefficient
 f : fugacity

$$G(p) = G^\ominus + nRT \ln \left(\frac{f}{p^\ominus} \right)$$

$$\ln \varphi = \int_0^p (Z - 1) \frac{dp}{p}$$

$Z < 1$, attractive interactions, $G_{\text{real}} < G_{\text{perfect}}$

$Z > 1$, repulsion interactions, $G_{\text{real}} > G_{\text{perfect}}$



1. 概念： 可逆过程、自发过程、 ΔS ΔA ΔG 在特定条件下与功和热的关系

2. Four criteria:

Engine efficiency ε

$\Delta S \geq 0$ *isolated system*

$\Delta A \leq 0$ *constant T, V ; $w = 0$*

$\Delta G \leq 0$ *constant T, p ; $w = 0$*

3. 计算 $\Delta S, \Delta G$



4.4 (a) Calculate the change in entropy when 25 kJ of energy is transferred reversibly and isothermally as heat to a large block of iron at (a) 0°C, (b) 100°C.

Isothermally $\Delta S = \int \frac{q_{rev}}{T} = \frac{1}{T} \int q_{rev} = \frac{q_{rev}}{T}$

Reversibly $q_{rev} = 25 \text{ kJ}$

$$\Delta S_1 = \frac{25 \text{ kJ}}{273 \text{ K}} = 91.58 \text{ J} \cdot \text{K}^{-1}$$

$$\Delta S_2 = \frac{25 \text{ kJ}}{373 \text{ K}} = 67.02 \text{ J} \cdot \text{K}^{-1}$$



4.10 (b) A system undergoes a process in which the entropy change is $+5.51 \text{ J K}^{-1}$. During the process, 1.50 kJ of heat is added to the system at 350 K . Is the process thermodynamically reversible? Explain your reasoning.

$$\sum \frac{dq}{T} = \frac{q}{T} = \frac{1.5 \text{ kJ}}{350 \text{ K}} = 4.29 \text{ J} \cdot \text{K}^{-1} < \Delta S$$

The process is not thermodynamically reversible



4.17 (b) The enthalpy of vaporization of methanol is $35.27 \text{ kJ mol}^{-1}$ at its normal boiling point of 64.1°C . Calculate (a) the entropy of vaporization of methanol at this temperature and (b) the entropy change of the surroundings.

$$\Delta_{vap}H = 35.27 \text{ kJ} \cdot \text{mol}^{-1}$$

$$\Delta_{vap}S = \frac{\Delta_{vap}H}{T} = \frac{35.27 \text{ kJ} \cdot \text{mol}^{-1}}{337.1 \text{ K}} = 104.6 \text{ J} \cdot \text{mol}^{-1}$$

At phase equilibrium $\Delta S_{iso} = \Delta S + \Delta S_{sur} = 0$

$$\Delta S_{sur} = -\Delta_{vap}S = -104.6 \text{ J} \cdot \text{mol}^{-1}$$



4.6 (a) A sample consisting of 1.00 mol of a monatomic perfect gas with $C_{V,m} = \frac{3}{2}R$ is heated from 100°C to 300°C at constant pressure. Calculate ΔS (for the system).

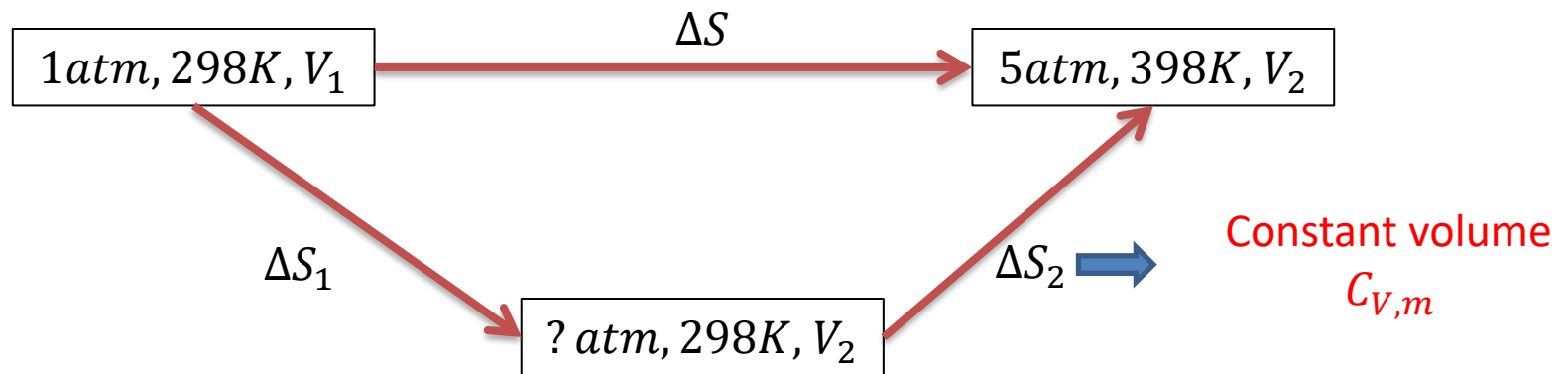
At constant pressure

$$\begin{aligned}\Delta S &= nC_{p,m} \ln \frac{T_f}{T_i} = n(C_{V,m} + R) \ln \frac{T_f}{T_i} \\ &= 2.5 \times 8.314 \times \ln \frac{573}{373} = 8.9 \text{ J} \cdot \text{K}^{-1}\end{aligned}$$



4.7 (a) Calculate ΔS (for the system) when the state of 3.00 mol of a monatomic perfect gas, for which $C_{p,m} = \frac{5}{2}R$, is changed from 25°C and 1.00 atm to 125°C and 5.00 atm. How do you rationalize the sign of ΔS ?

$$1 \text{ atm} \times \frac{V_1}{nR \times 298K} = 5 \text{ atm} \times \frac{V_2}{nR \times 398K} \Rightarrow \frac{V_2}{V_1} = 0.27$$



$$\Delta S = \Delta S_1 + \Delta S_2 = nR \ln \frac{V_2}{V_1} + n(C_{p,m} - R) \ln \frac{T_f}{T_i}$$

$$= 3 \times 8.314 \times \ln 0.27 + 3 \times 1.5 \times 8.314 \times \ln 1.34 = -21.8 \text{ J} \cdot \text{K}^{-1}$$

$$nR \ln \frac{V_f}{V_i} = nR \ln \frac{p_i}{p_f}$$



4.8 (a) A sample consisting of 3.00 mol of a diatomic perfect gas at 200 K is compressed reversibly and adiabatically until its temperature reaches 250 K. Given that $C_{V,m} = 27.5 \text{ J K}^{-1} \text{ mol}^{-1}$, calculate q , w , ΔU , ΔH , and ΔS .

3 mol, 200 K

Compressed reversibly
Adiabatically $q=0$

3 mol, 250 K

$$q_{rev} = 0$$

$$\Delta S = 0$$

$$\Delta U = nC_{V,m}\Delta T = 3 \times 27.5 \times 50 = 4.125 \text{ kJ}$$

$$\Delta H = n(C_{V,m} + R)\Delta T = 3 \times (27.5 + 8.314) \times 50 = 5.372 \text{ kJ}$$

$$\Delta U = q_{rev} + w \quad \longrightarrow \quad w = \Delta U = 4.125 \text{ kJ}$$



4.11 (b) A sample of copper of mass 2.75 kg is cooled at constant pressure from 330 K to 275 K. Calculate (a) the energy that must be removed as heat and (b) the change in entropy of the sample.

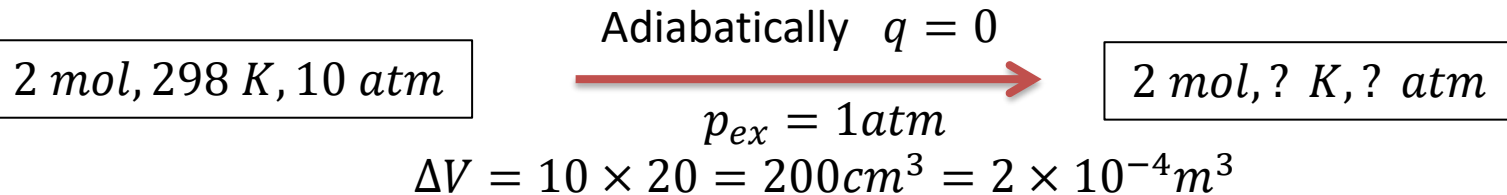
$$C_{p,m}^{\ominus}(\text{copper}) = 24.44 \text{ J} \cdot \text{K}^{-1} \text{ mol}^{-1}$$

$$q = nC_{p,m}\Delta T = \frac{2750}{63.5} \times 24.44 \times (-55) = -58.2 \text{ kJ}$$

$$\begin{aligned}\Delta S &= nC_{p,m} \ln \frac{T_f}{T_i} = \frac{2750}{63.5} \times 24.44 \times \ln \frac{275}{330} \\ &= -193.0 \text{ J} \cdot \text{K}^{-1}\end{aligned}$$



4.16 (a) Consider a system consisting of 2.0 mol $\text{CO}_2(\text{g})$, initially at 25°C and 10 atm and confined to a cylinder of cross-section 10.0 cm^2 . It is allowed to expand adiabatically against an external pressure of 1.0 atm until the piston has moved outwards through 20 cm. Assume that carbon dioxide may be considered a perfect gas with $C_{V,m} = 28.8\text{ J K}^{-1}\text{ mol}^{-1}$ and calculate (a) q , (b) w , (c) ΔU , (d) ΔT , (e) ΔS .

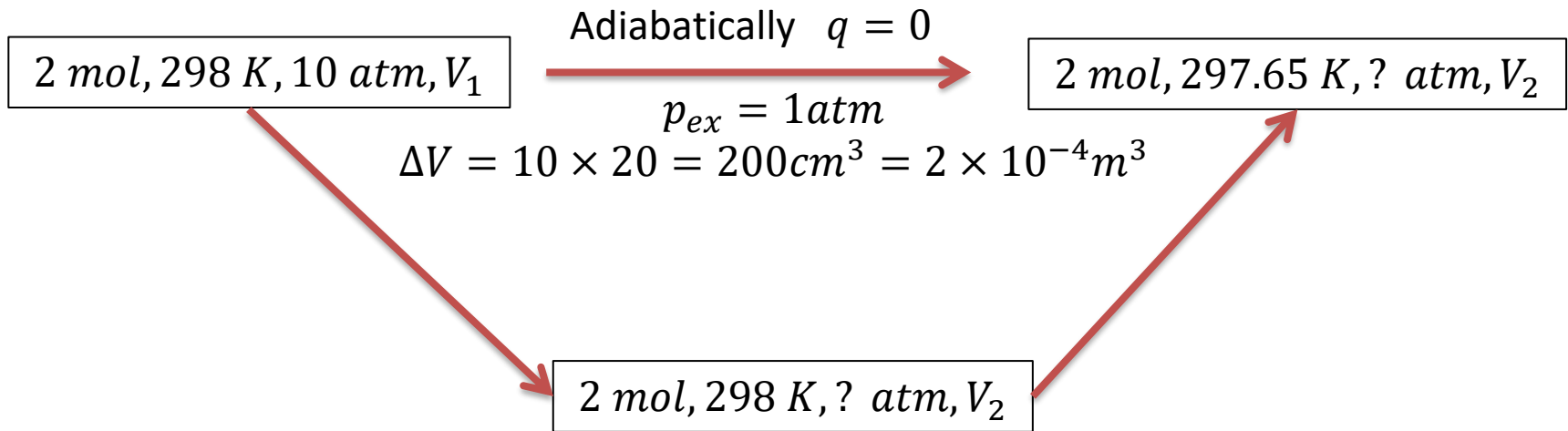


$$q = 0$$

$$w = -p_{\text{ex}} \Delta V = -1.013 \times 10^5\text{ Pa} \times 2 \times 10^{-4}\text{ m}^3 = -20\text{ J}$$

$$\Delta U = q + w = -20\text{ J}$$

$$\Delta U = nC_{V,m}\Delta T = 2 \times 28.8 \times \Delta T = -20\text{ J} \quad \Rightarrow \quad \Delta T = -0.35\text{ K}$$



$$V_1 = \frac{2 \times 8.314 \times 298}{10 \times 1.01 \times 10^5} = 4.906 \times 10^{-3} \text{ m}^3$$

$$V_2 = V_1 + \Delta V = 4.906 \times 10^{-3} + 2 \times 10^{-4} = 5.106 \times 10^{-3} \text{ m}^3$$

$$\begin{aligned} \Delta S &= \Delta S_1 + \Delta S_2 = nR \ln \frac{V_2}{V_1} + nC_{V,m} \ln \frac{T_f}{T_i} \\ &= 2 \times 8.314 \times \ln \frac{5.106}{4.906} + 2 \times 28.8 \times \ln \frac{297.65}{298} = 0.6 \text{ J} \cdot \text{K}^{-1} \end{aligned}$$



4.23 (a) Calculate the change in the entropies of the system and the surroundings, and the total change in entropy, when a sample of nitrogen gas of mass 14 g at 298 K and 1.00 bar doubles its volume in (a) an isothermal reversible expansion, (b) an isothermal irreversible expansion against $p_{\text{ex}} = 0$, and (c) an adiabatic reversible expansion.

14g (0.5 mol), 298K, 1 bar, V

14g (0.5 mol), 298K, ? bar, $2V$

(a) Isothermal reversible

$$\Delta U = 0$$

$$\Delta S = nR \ln \frac{V_f}{V_i} = 0.5 \times 8.314 \times \ln 2 = 2.88 \text{ J} \cdot \text{K}^{-1}$$

Reversible process

$$\Delta S_{\text{iso}} = \Delta S + \Delta S_{\text{sur}} = 0$$

$$\Delta S_{\text{sur}} = -\Delta_{\text{vap}} S = -2.88 \text{ J} \cdot \text{mol}^{-1}$$

14g (0.5 mol), 298K, 1 bar, V

14g (0.5 mol), 298K, ? bar, $2V$

(b) Isothermal irreversible $p_{ex} = 0$

因为恒温，且前后体积变化相同
可判定初始状态和 (a) 问相同

ΔS is a state function

$$\Delta S = 2.88 J \cdot K^{-1}$$

$$w = 0 \quad \Delta U = 0 \quad \Rightarrow \quad q = 0$$

$$\Delta S_{sur} = 0$$

$$\Delta S_{iso} = \Delta S + \Delta S_{sur} = 2.88 J \cdot K^{-1}$$

14g (0.5 mol), 298K, 1 bar, V

14g (0.5 mol), ? K, ? bar, 2V

(c) Adiabatic reversible

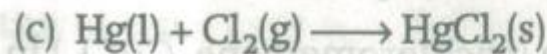
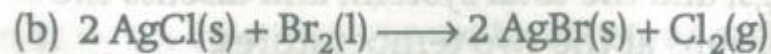
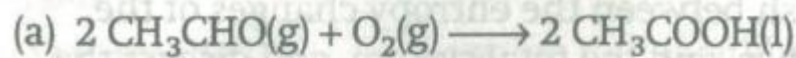
$$q_{rev} = 0$$

$$\Delta S = 0$$

$$\Delta S_{sur} = 0$$

$$\Delta S_{iso} = \Delta S + \Delta S_{sur} = 0$$

4.20 (a) Use standard Gibbs energies of formation to calculate the standard reaction Gibbs energies at 298 K of the reactions in Exercise 4.18a.



20 a (a) $\Delta_r G^\ominus = 2\Delta_f G^\ominus(\text{CH}_3\text{COOH}, \text{l}) - 2\Delta_f G^\ominus(\text{CH}_3\text{CHO}, \text{g})$
 $= -2 \times 389.9 + 2 \times 128.86$
 $= -522.08 \text{ kJ/mol}$

(b) $\Delta_r G^\ominus = 2\Delta_f G^\ominus(\text{AgBr}, \text{s}) - 2\Delta_f G^\ominus(\text{AgCl}, \text{s}) - \Delta_f G^\ominus(\text{Br}_2, \text{l})$
 $= 2 \times (-96.90) - 2 \times (-109.79) - 3.110$
 $= 25.78 \text{ kJ/mol}$

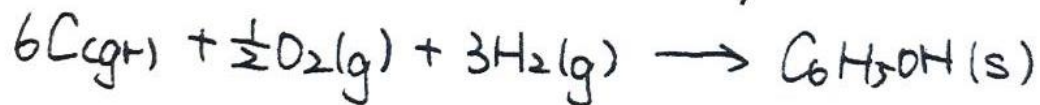
(c) $\Delta_r G^\ominus = \Delta_f G^\ominus(\text{HgCl}_2, \text{s})$
 $= -178.6 \text{ kJ/mol}$

4.22 (a) The standard enthalpy of combustion of solid phenol ($\text{C}_6\text{H}_5\text{OH}$) is $-3054 \text{ kJ mol}^{-1}$ at 298 K and its standard molar entropy is $144.0 \text{ J K}^{-1} \text{ mol}^{-1}$. Calculate the standard Gibbs energy of formation of phenol at 298 K.

22 a The combustion reaction:



$$\begin{aligned}\Delta_f H^\ominus(\text{C}_6\text{H}_5\text{OH}, \text{s}) &= 6\Delta_f H^\ominus(\text{CO}_2, \text{g}) + 3\Delta_f H^\ominus(\text{H}_2\text{O}, \text{l}) - \Delta_c H(\text{C}_6\text{H}_5\text{OH}) \\ &= 6 \times (-393.5) + 3 \times (-285.83) - (-3054) \\ &= -164.55 \text{ kJ/mol}\end{aligned}$$



$$\begin{aligned}\Delta_f S^\ominus &= S_m^\ominus(\text{C}_6\text{H}_5\text{OH}) - 6S_m^\ominus(\text{C}, \text{gr}) - \frac{1}{2}S_m^\ominus(\text{O}_2, \text{g}) - 3S_m^\ominus(\text{H}_2, \text{g}) \\ &= 144.0 - 6 \times (5.740) - \frac{1}{2}(205.138) - 3 \times (130.684) \\ &= -385.06 \text{ J K}^{-1} \text{ mol}^{-1}\end{aligned}$$

$$\Delta_f G^\ominus = \Delta_f H^\ominus - T\Delta_f S^\ominus$$

$$\begin{aligned}&= -164.55 \text{ kJ/mol} - (298 \text{ K}) \times (-385.061) \text{ J/mol} \\ &= -49.80 \text{ kJ/mol}\end{aligned}$$

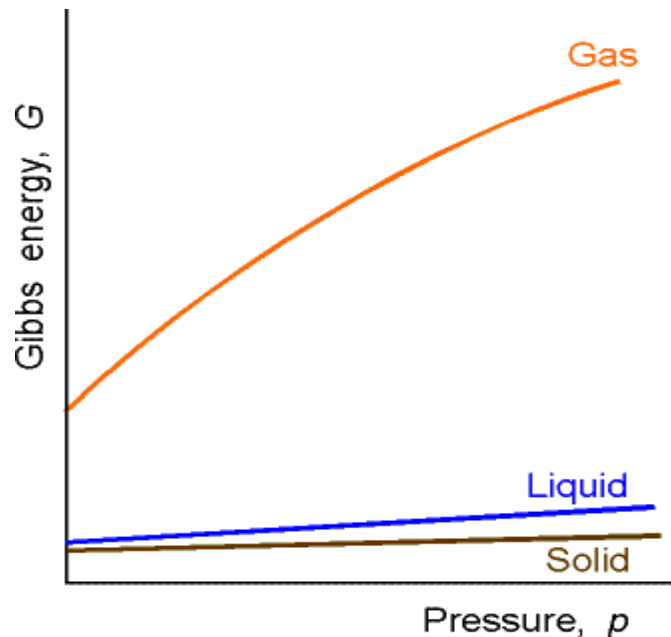
5.6 (a) The change in the Gibbs energy of a certain constant-pressure process was found to fit the expression $\Delta G/\text{J} = -85.40 + 36.5(T/\text{K})$. Calculate the value of ΔS for the process.

$$dG = Vdp - SdT \xrightarrow{\text{Constant Pressure}} \left(\frac{\partial G}{\partial T}\right)_p = -S$$

5.6a

$$\begin{aligned}\left(\frac{\partial G}{\partial T}\right)_p &= -S & \left(\frac{\partial G_f}{\partial T}\right)_p &= -S_f & \left(\frac{\partial G_i}{\partial T}\right)_p &= -S_i \\ \Delta S = S_f - S_i &= -\left(\frac{\partial G_f}{\partial T}\right)_p + \left(\frac{\partial G_i}{\partial T}\right)_p \\ &= -\left\{\frac{\partial (G_f - G_i)}{\partial T}\right\}_p \\ &= -\left(\frac{\partial \Delta G}{\partial T}\right)_p \\ &= -\frac{d}{dT}(-85.40 + 36.5T) \\ &= -36.5 \text{ J K}^{-1}\end{aligned}$$

5.7 (a) Calculate the change in Gibbs energy of 35 g of ethanol (mass density 0.789 g cm^{-3}) when the pressure is increased isothermally from 1 atm to 3000 atm.



$$\Delta G = \Delta p \cdot V$$

5.7a

$$\Delta G = V \Delta p \quad V = \frac{m}{\rho} = \frac{35\text{g}}{0.789} = 44.36 \text{ cm}^3$$

$$\Delta G = 44.36 \times 10^{-6} \times 2999 \times 101325$$

$$= 13.48 \text{ kJ}$$

The variation of the Gibbs energy with the pressure determined by the volume of the sample

Liquid or solid is not sensitive to pressure change

Almost linear

$$\left(\frac{\partial G_m}{\partial p}\right)_T = V_m \quad \Rightarrow \quad \Delta G_m = \Delta p \cdot V_m$$

5.8 (a) When 2.00 mol of a gas at 330 K and 3.50 atm is subjected to isothermal compression, its entropy decreases by 25.0 J K^{-1} . Calculate (a) the final pressure of the gas and (b) ΔG for the compression.

Isothermal compression

$$\Delta S = nR \ln \left(\frac{V_f}{V_i} \right) = nR \ln \left(\frac{p_i}{p_f} \right) = -25 \text{ J/K}$$

$$p_f = p_i e^{-\Delta S/nR} = 3.5 \text{ atm} \times e^{25/2 \times 8.314} = 15.7 \text{ atm}$$

$$\Delta G = \Delta H - T\Delta S = -T\Delta S = -330 \text{ K} \times \left(-\frac{25 \text{ J}}{\text{K}} \right) = 8.25 \text{ kJ}$$

5.9 (a) Calculate the change in molar Gibbs energy of a perfect gas when its pressure is increased isothermally from 1.8 atm to 29.5 atm at 40°C.

$$\Delta G_m = RT \ln \left(\frac{p_f}{p_i} \right) = 8.314 \times 313 \times \ln \left(\frac{29.5}{1.8} \right) \\ = 7.3 \text{ kJ/mol}$$

5.15 (a) Evaluate $(\partial S/\partial V)_T$ for a van der Waals gas. For an isothermal expansion, will ΔS be greater for a perfect gas or a van der Waals gas? Explain your conclusion.

According to the Maxwell's Relation
$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$$

For van der Waals gas:
$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V = \frac{R}{V_m - b} \quad \Delta S_{van\ der} = \int \frac{R}{V_m - b} dV_m = R \ln \frac{V_{m,f} - b}{V_{m,i} - b}$$

For perfect gas:
$$\Delta S_{per} = R \ln \frac{V_{m,f}}{V_{m,i}}$$

$$\Delta S_{van\ der} - \Delta S_{per} = R \ln \frac{V_{m,f} \cdot V_{m,i} - bV_{m,i}}{V_{m,f} \cdot V_{m,i} - bV_{m,f}}$$

$\therefore$ isothermal expansion $V_{m,i} < V_{m,f}$

$$\therefore \frac{V_{m,f} \cdot V_{m,i} - bV_{m,i}}{V_{m,f} \cdot V_{m,i} - bV_{m,f}} > 1 \quad \Delta S_{van\ der} - \Delta S_{per} > 0$$

$\Delta S_{van\ der} > \Delta S_{per}$ ΔS is greater for van der Waals gas.